

## CONJURA · OPEN PROBLEM

# *The Open Generalized Mirror Theory Conjecture*

*Exact Multi-Graph Bounds Beyond the  $O(2^{n/4})$  Regime*

**Status.** Statement: AI-written, not yet formalized.

**Category.** Symmetric-Key Cryptography.

Mirror Theory, initially introduced by [Patarin \(2003\)](#), provides lower bounds on the number of non-degenerate solutions to systems of bivariate linear equations over  $\mathbb{F}_2^n$ . While recent breakthroughs (e.g., [Cogliati et al., 2023](#)) have rigorously established the theory for component sizes up to  $\xi_{\max} = O(2^{n/4}/\sqrt{n})$ , the fully generalized conjecture for arbitrary dependency graphs up to the information-theoretic limit remains an open problem in theoretical cryptography. This document provides a self-contained statement of the open Generalized Mirror Theory Conjecture.

## 1 Formal Preliminaries

Let  $V = \{p_1, p_2, \dots, p_q\}$  be a set of variables taking values in  $\{0, 1\}^n$ , where  $N = 2^n$ . We consider systems of bivariate affine equations over  $\mathbb{F}_2^n$  defined on a multi-graph  $G = (V, E)$ .

**Definition 1 (Affine Equation System).** Let  $G = (V, E)$  be a connected graph with  $q = |V|$  vertices and  $m = |E|$  edges. Let  $\lambda : E \rightarrow \{0, 1\}^n \setminus \{0^n\}$  be an edge-labeling function assigning non-zero target constants to each edge. The system of equations is given by:

$$\mathcal{E}(G, \lambda) := \left\{ p_i \oplus p_j = \lambda_{i,j} \mid e = (p_i, p_j) \in E \right\}. \quad (1)$$

**Definition 2 (Non-Degenerate Solutions).** A tuple  $(P_1, \dots, P_q) \in (\{0, 1\}^n)^q$  is called a **valid (non-degenerate) solution** to  $\mathcal{E}(G, \lambda)$  if:

- i.  $P_i \oplus P_j = \lambda_{i,j}$  for all  $(p_i, p_j) \in E$ .
- ii.  $P_i \neq P_j$  for all  $i \neq j$  (all variables in  $V$  are pairwise distinct).

We denote the set of all non-degenerate solutions by  $\mathbf{Sol}(\mathcal{E}(G, \lambda))$  and its cardinality by  $h(G, \lambda) := |\mathbf{Sol}(\mathcal{E}(G, \lambda))|$ .

**Definition 3 (Cycle Consistency and Component Size).**

A system  $\mathcal{E}(G, \lambda)$  is **cycle-consistent** if for every cycle  $C$  in  $G$ , the XOR sum of labels along the cycle evaluates to  $0^n$ . Let  $\xi_{\max}(G)$  denote the maximum size (number of vertices) of any connected component of  $G$ , and let  $\nu(G) = |E| - |V| + c(G)$  be the circuit rank (cyclomatic number) of  $G$ , where  $c(G)$  is the number of connected components.

## 2 The Unconditional Generalized Mirror Theory Conjecture

While localized versions of Mirror Theory have been proven for trees and bounded-degree graphs up to  $\xi_{\max} = O(N^{1/4})$ , the full asymptotic statement for arbitrary graph topologies up to  $O(N^{1/2})$  or higher query limits remains open.

**Conjecture 1 (Generalized Mirror Theory Conjecture).** *Let  $G = (V, E)$  be a graph with  $q = |V|$  vertices and  $m = |E|$  edges. Let  $\mathcal{E}(G, \lambda)$  be a cycle-consistent affine system. Assume that  $q \leq o(N^{1/2})$  and that the maximum component size satisfies  $\xi_{\max}(G) \leq o(N^{1/2})$ . Then, the number of non-degenerate solutions  $h(G, \lambda)$  satisfies the tight lower bound:*

$$h(G, \lambda) \geq \frac{(N)_q}{N^m} \cdot \left( 1 - O\left( \frac{q \cdot \xi_{\max}(G)}{N} \right) \right), \quad (2)$$

where  $(N)_q = N(N-1) \dots (N-q+1)$  denotes the falling factorial. Equivalently, for a collection of  $c$  connected tree components  $T_1, \dots, T_c$

with  $|V(T_k)| = q_k$ ,  $q = \sum q_k$ , and  $m = q - c$ :

$$h(G, \lambda) \geq \frac{N^q}{N^{q-c}} \prod_{i=0}^{q-1} \left(1 - \frac{i}{N}\right) (1 - \epsilon(q, \xi_{\max}, N)), \quad (3)$$

where  $\epsilon(q, \xi_{\max}, N) \rightarrow 0$  as long as  $q \cdot \xi_{\max}(G) = o(N)$ .

### 3 State of the Art and Theoretical Gaps

| Reference                         | Scope / Topology              | Max Component ( $\xi_{\max}$ )     | Status          |
|-----------------------------------|-------------------------------|------------------------------------|-----------------|
| Patarin (2003),<br>Patarin (2005) | Linear Graphs / Trees         | $\xi_{\max} = O(1)$                | Gaps identified |
| Cogliati &<br>Seurin (2018)       | Circle Graphs ( $\nu = 1$ )   | $\xi_{\max} = O(N^{1/3})$          | Proven          |
| Cogliati et al.<br>(2023)         | Arbitrary Forest Graphs       | $\xi_{\max} = O(N^{1/4}/\sqrt{n})$ | Fully Proven    |
| <b>Open Conjecture</b>            | <b>Arbitrary Multi-Graphs</b> | $\xi_{\max} = o(N^{1/2})$          | <b>OPEN</b>     |

Table 1: Evolution of Mirror Theory bounds for system size  $q$  and component size  $\xi_{\max}$ .

#### Why the Conjecture Remains Open:

1. **High-Order Link Deletions:** Current algebraic techniques (such as link-deletion equations) rely on induction over component sizes. When  $\xi_{\max}$  exceeds  $O(N^{1/4})$ , higher-order dependency intersections generate non-trivial error terms that cannot be absorbed without explicit structural cancellation lemmas.
2. **Multi-Cycle Dependencies:** For graphs with high cyclomatic numbers ( $\nu(G) > 1$ ), handling non-equations ( $P_i \neq P_j$ ) alongside linear constraints requires tracking exact character sums over non-abelian representations, which currently lack tight lower bounds.